

Def.

A deformation retraction of a space X onto a subspace A is a family

$\{f_t: X \rightarrow X \mid t \in I\}$ satisfying $f_0 \equiv \text{id}$, $f_1[X] = A$, $f_t|_A \equiv \text{id}_A$ for all t , and $X \times I \rightarrow X: (x, t) \mapsto f_t(x)$ is continuous.

More generally, define

Def.

A Homotopy of a space X into a space Y is a family

$\{f_t: X \rightarrow Y \mid t \in I\}$ satisfying the associated map $F: X \times I \rightarrow Y: (x, t) \mapsto f_t(x)$ is continuous.

Moreover, given two maps g_0, g_1 are called homotopic if there is a homotopy. In such a situation, denote $g_0 \simeq g_1$.

Recall that: a map $r: X \rightarrow X$ satisfying $r[X] = A$ and $r|_A \equiv \text{id}_A$ is called retraction. Thus, we can say that the deformation retraction of a space X onto a subspace A is a homotopy from the identity map of X to a retraction of X onto A . And, algebraically, the retraction $r: X \rightarrow X$ satisfies that $r \circ r = r^2 = r$, analogous to the projection operator.

Def.

A map $f: X \rightarrow Y$ is called a homotopy equivalence if: there exists a map $g: Y \rightarrow X$ s.t.
 $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$.

If such a map exists, then the spaces X and Y are called homotopy equivalent.

Def.

A space that homotopy equivalent to a point is called contractible.

